

Policy Gradient Methods + Actor Critic Algorithms (Chapter 13)

- Here we consider approaches that are parameterizing the policy.
- Let $\pi(a|s, \theta) = P(A_t = a | S_t = s, \theta)$
where $\theta \in \mathbb{R}^{d'}$ is the policy parameter.
- The performance objective used to optimize θ may or may not be the value function.
- If value function is considered,

we will let it's weight be

$$w \in \mathbb{R}^d$$

- Let $J: \mathbb{R}^{d'} \rightarrow \mathbb{R}$ be the performance objective to be optimized

- Algorithm:

$$\theta_{t+1} = \theta_t + \alpha \hat{\nabla J}(\theta_t)$$

stochastic estimate

$$\nabla J(\theta_t)$$

s.t.

$$E[\hat{\nabla J}(\theta_t)] = \nabla J(\theta_t)$$

- These algorithms are called policy

Gradient (PG) algorithms.

- Actor-Critic Algorithms: Methods that learn to approximate both value functions and policy.


- Assumption (on policy):

$$\pi(a|s, \theta) > 0 \quad \forall a, s, \theta.$$

Moreover, $\pi(a|s, \theta)$ is differentiable w.r.t. θ and $\nabla_{\theta} \pi(a|s, \theta)$ exists and is finite $\forall s, a, \theta$

Example of policy parametrization

Let $h(s, a, \theta) \in \mathbb{R}$ denote the numerical preference for each state-action pair.

$$\text{Let } \pi(a|s, \theta) = \frac{e^{h(s, a, \theta)}}{\sum_{b \in A(s)} e^{h(s, b, \theta)}}$$


Soft max
policies, or

Parameterized Boltzmann
policies

Examples of $h(s, a, \theta)$:

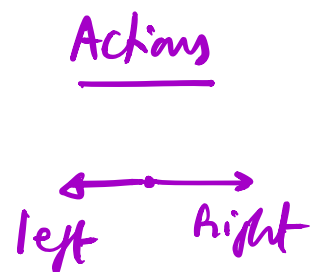
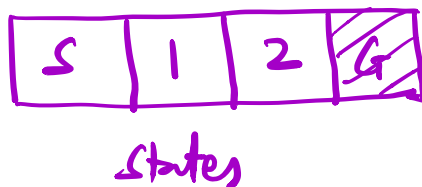
(i) linear parametrization

$$h(s, a, \theta) = \theta^T \phi(s, a)$$

(ii) $h(s, a, \theta)$ may be computed by
a deep artificial neural network
(e.g., used in AlphaGo)

Example [why policy parametrization
might be beneficial]

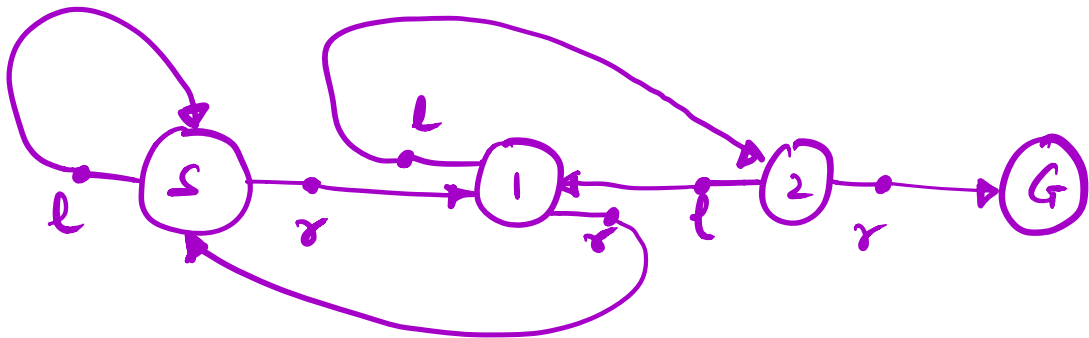
Small Grid world



Reward = -1 per step

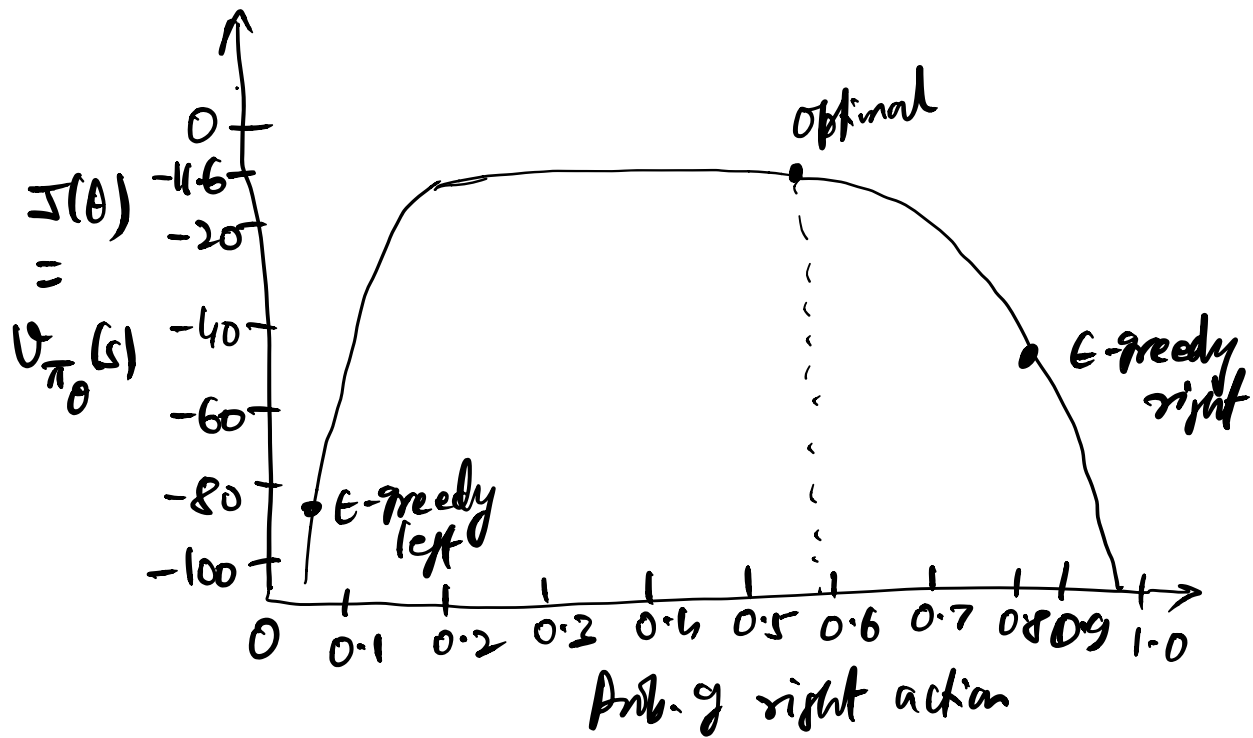
$s, 1, 2$: nonterminal states

G : goal state



Typically, an action value method with ϵ -greedy action selection is forced to choose between two policies

- (i) Choose right w.p. $1 - \frac{\epsilon}{2}$ on all steps
- (ii) Choose left w.p. $1 - \frac{\epsilon}{2}$ — . . .



If $\epsilon = 0.1$, then value of ϵ -greedy left at start state $s = -82$

----- right -----
 ----- = -44

* A method can do better if it learns the right prob. to pick the right action

* Best or optimal probability = 0.594

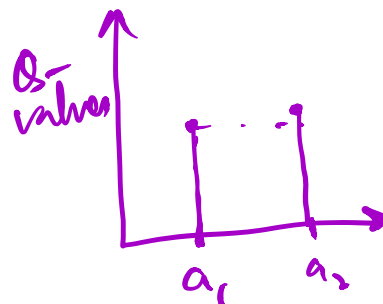
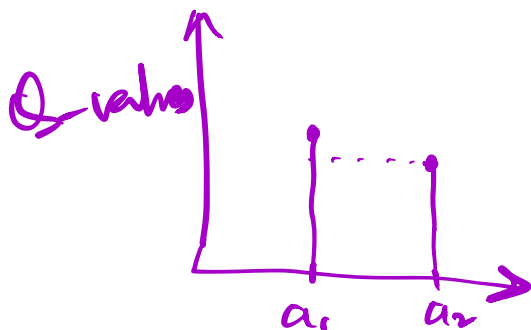
value achieved = -11.6

Advantage of policy parameterization over ϵ -greedy action selection:

* A small change in parameter

$\Rightarrow$ small change in action selection probability

* In ϵ -greedy case, small change in action value $\Rightarrow$ different action being chosen



Actions at time t	Actions at time $t+1$
a_1 is greedy action	a_2 is greedy action
selected w.p. $1-\epsilon$	selected w.p. $1-\epsilon$

- Stronger convergence guarantees are available for PG methods since policies are smoother.

Consider the episodic case. Let

$$J(\theta) = V_{\pi_\theta}(s_0) \quad \text{where } s_0$$

is the start state (non random)

U_{π_θ} : value function for π_θ .

Assume $\gamma = 1$

Policy Gradient Theorem:

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a q_{\pi}(s,a) \nabla_{\theta} \pi(a|s,\theta)$$

Here $\mu(s) \equiv$ av. fraction of time
that Markov chain spends in
state s .

Notice: $J(\theta) = U_{\pi_\theta}(s_0)$

$$\therefore \nabla J(\theta) = \nabla U_{\pi_\theta}(s_0)$$

$$\nabla V_{\pi_{\theta}}(s) = \nabla \left(\sum_a \pi(a|s) q_{\pi}(s, a) \right) \quad \forall s \in \mathcal{S}$$

$$= \sum_a \left(\nabla \pi(a|s) q_{\pi}(s, a) + \pi(a|s) \nabla q_{\pi}(s, a) \right)$$

$$= \sum_a \left[\nabla \pi(a|s) q_{\pi}(s, a) + \pi(a|s) \nabla \left(\sum_{s', r} p(s', r|s, a) * (r + V_{\pi}(s')) \right) \right]$$

$$= \sum_a \nabla \pi(a|s) q_{\pi}(s, a) + \pi(a|s) \sum_{s', r} p(s', r|s, a) \nabla V_{\pi}(s')$$

$$= \sum_a \nabla \pi(a|s) q_{\pi}(s, a)$$

$$+ \pi(a|s) \sum_{s'} \bar{p}(s'|s, a) \nabla U_{\pi}(s')$$

$$\text{where } \bar{p}(s'|s, a) = \sum_r p(s', r|s, a)$$

Unfolding the above,

$$= \sum_a \left[\nabla \pi(a|s) q_{\pi}(s, a) \right.$$

$$+ \pi(a|s) \sum_{s'} \bar{p}(s'|s, a) \left[\sum_{a'} \left(\nabla \pi(a'|s') q_{\pi}(s', a') \right. \right. \\ \left. \left. + \pi(a'|s') \sum_{s''} \bar{p}(s''|s', a') * \nabla U_{\pi}(s'') \right) \right]$$

Upon repeated unfolding, we obtain

$$\nabla V_{\pi}(s) = \sum_{x \in S} \sum_{k=0}^{\infty} P(s \rightarrow x, k, \pi) * \sum_a \nabla \pi(a|x) V_{\pi}(x, a)$$

Here $P(s \rightarrow x, k, \pi) \equiv$ prob. of going from state s to state x in k steps under π .

Thus,

$$\nabla V_{\pi}(s) = \sum_x \eta(x) \sum_a \nabla \pi(a|x) V_{\pi}(x, a)$$

where $\eta(x) = \sum_{k=0}^{\infty} P(s_0 \rightarrow x, k, \pi)$



$$= \sum_{s'} \eta(s') \sum_x \underbrace{\frac{\eta(x)}{\sum_{s'} \eta(s')}}_{\mu(x)} \sum_a \nabla \pi(a|x) q_{\pi}(x,a)$$

$$= \sum_{s'} \eta(s') \sum_x \mu(x) \sum_a \nabla \pi(a|x) q_{\pi}(x,a)$$

$$\propto \sum_x \mu(x) \sum_a \nabla \pi(a|x) q_{\pi}(x,a) \rightarrow (1)$$

Here μ is the state distribution under π .

REINFORCE: Monte-Carlo PG Algorithm

Idea: Since we don't know $q_{\pi}(s, a)$ in (1), we need to estimate it.

Now

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a q_{\pi}(s, a) \nabla_{\theta} \pi(a|s, \theta)$$

$$= E_{\pi} \left[\sum_a q_{\pi}(s_t, a) \nabla_{\theta} \pi(a|s_t) \right]$$

$$= E_{\pi} \left[\sum_a \pi(a|s_t) q_{\pi}(s_t, a) \frac{\nabla_{\theta} \pi(a|s_t)}{\pi(a|s_t)} \right]$$

$$= E_{\pi} \left[q_{\pi}(s_t, A_t) \frac{\nabla \pi(A_t | s_t)}{\pi(A_t | s_t)} \right]$$

where $A_t \sim \pi$

$$= E_{\pi} \left[G_t \frac{\nabla \pi(A_t | s_t)}{\pi(A_t | s_t)} \right]$$

where $G_t \equiv$ return from time t

REINFORCE (stochastic gradient ascent
update)

$$\theta_{t+1} = \theta_t + \alpha G_t \frac{\nabla \pi(A_t | s_t)}{\pi(A_t | s_t)}$$

→ (2)

Note: REINFORCE is a Monte-Carlo algorithm (uses complete return from time t)

REINFORCE with Baseline:

Let $b: \mathcal{S} \rightarrow \mathbb{R}$ be a certain function.

Such a function is called the Baseline function.

The PG theorem can be generalized to

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a (r_{\pi}(s, a) - b(s)) \nabla \pi(a|s)$$

Note: $\sum_a b(s) \nabla \pi(a|s)$

$$= b(s) \sum_a \nabla \pi(a|s)$$

$$= b(s) \nabla \left(\sum_a \pi(a|s) \right)$$

$$= b(s) \nabla(1) = 0$$

$$\left(\text{since } \sum_a \pi(a|s) = 1 \right)$$

* Advantage: With a suitable choice of

$b(s)$, one may be able to significantly reduce the variance of the estimates

* It has been shown that value function

$V_{\pi_\theta}(s)$ as the baseline significantly reduces variance of PG estimator.

Recall: REINFORCE update has the form

$$\theta_{t+1} = \theta_t + \alpha G_t \frac{\nabla \pi(A_t | s_t, \theta_t)}{\pi(A_t | s_t, \theta_t)}$$

- REINFORCE with baseline update has the form

$$\theta_{t+1} = \theta_t + \alpha (G_t - b(s_t)) \frac{\nabla \pi(A_t | s_t, \theta_t)}{\pi(A_t | s_t, \theta_t)}$$

To estimate value function, one may

use value function parameters

REINFORCE with Baseline (Episodic), for

estimating $\pi_\theta \approx \pi^*$

Input: A differentiable policy parametrization
 $\pi(a|s, \theta)$

Input: A differentiable state-value function
parametrization $\hat{v}(s, w)$

Algorithm parameters: step sizes $\alpha^\theta > 0$, $\alpha^w > 0$

Initialize policy parameter $\theta \in \mathbb{R}^{d'}$ and
state value weights $w \in \mathbb{R}^d$ (e.g., to 0)

- Loop forever (for each episode):

Generate an episode $s_0, A_0, R_1, s_1, A_1,$
 $R_2, s_2, \dots, s_{T-1}, A_{T-1}, R_T, s_T$

following $\pi(\cdot | \cdot, \theta)$

- Loop for each step of episode

$t = 0, 1, \dots, T-1$:

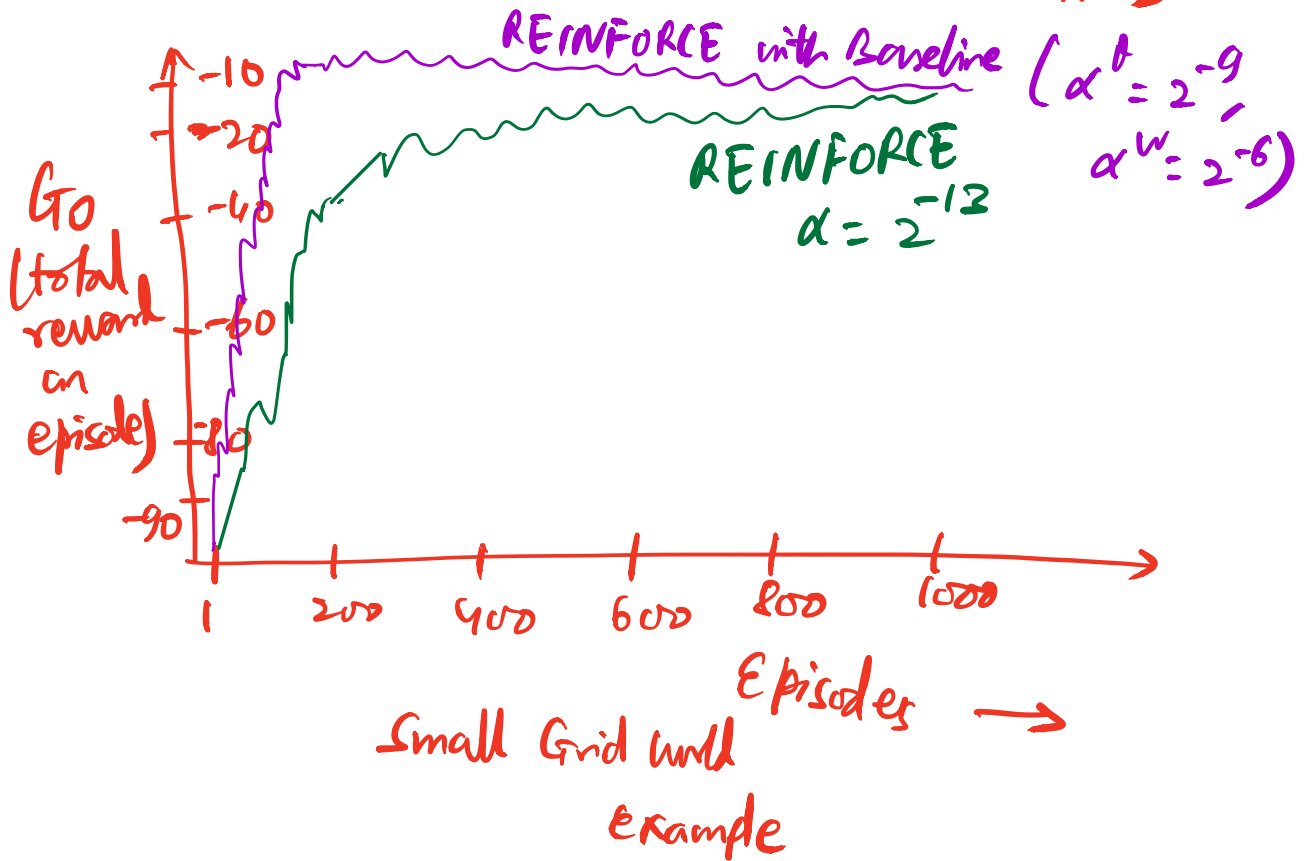
$$G \leftarrow \sum_{k=t+1}^{T-1} \gamma^{k-t-1} R_k$$

$$\delta \leftarrow G - \hat{v}(s_t, w)$$

$$w \leftarrow w + \alpha^w \delta \nabla \hat{v}(s_t, w)$$

$$\theta \leftarrow \theta + \alpha^{\theta} \delta \nabla \pi(A_t | s_t, \theta)$$

$$\pi(A_t | s_t, \theta)$$



Actor-Critic Methods:

In one-step actor-critic methods, we replace the full return of REINFORCE

with the one-step return along with the value function as baseline.

Policy parameter update:

$$\begin{aligned}\theta_{t+1} &= \theta_t + \alpha \left(G_{t:t+1} - \hat{v}(s_t, \omega) \right) \frac{\nabla \pi(A_t | s_t, \theta_t)}{\pi(A_t | s_t, \theta_t)} \\ &= \theta_t + \alpha \left(R_{t+1} + \gamma \hat{v}(s_{t+1}, \omega) - \hat{v}(s_t, \omega) \right) * \\ &\quad \frac{\nabla \pi(A_t | s_t, \theta_t)}{\pi(A_t | s_t, \theta_t)}\end{aligned}$$

One-step Actor-Critic (episodic), for estimating

$$\pi_{\theta} \approx \pi_{*}:$$

Input: a differentiable policy parametrization
 $\pi(a|s, \theta)$

Input: a differentiable state-value function
parametrization $\hat{V}(s, w)$

Parameters: step sizes $\alpha^\theta > 0, \alpha^w > 0$

- Initialize policy parameter $\theta \in \mathbb{R}^{d'}$ and
state-value function weight $w \in \mathbb{R}^d$ (e.g., to
0)
- Loop forever (for each episode):
 - $I \leftarrow 1$
 - Loop while $\mathcal{I}$ is not terminal

(for each time step)

- $A \sim \pi(\cdot | s, \theta)$
- Take action A , observe s', R
- $\delta \leftarrow R + \gamma \hat{V}(s', \omega) - \hat{V}(s, \omega)$
[if s' is terminal, then set $\hat{V}(s', \omega) = 0$]
- $W \leftarrow W + \alpha^W \delta \nabla \hat{V}(s, \omega)$
- $\theta \leftarrow \theta + \alpha^\theta \mathbb{I} \delta \nabla \ln \pi(A | s, \theta)$

$$\left| \begin{aligned} &\nabla \ln \pi(A | s, \theta) \\ &= \frac{\nabla \pi(A | s, \theta)}{\pi(A | s, \theta)} \end{aligned} \right|$$

$$- \quad I \leftarrow \gamma I$$

$$- \quad s \leftarrow s'$$

* Important to note:

- In actor-critic algorithms, value function update is performed according to TD learning (for a specified policy)
- Policy update is done using policy gradient algorithm
- Actor-critic algorithm thus

updates both the policy & value function.

Policy Gradient Methods for Continuous Problems:

- Infinite horizon discounted reward setting

Let

$$V_{\pi}(s) = E \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s, \pi \right]$$

$$Q_{\pi}(s, a) = E \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s, A_0 = a, \pi \right]$$

Here $\gamma \in (0, 1)$

Policy Gradient Theorem

$$\nabla U_{\pi}(s) = \sum_s d^{\pi}(s) \sum_a \nabla \pi(a|s) Q^{\pi}(s,a) \quad \forall s \in \mathcal{S}$$

Note:
$$U^{\pi}(s_0) = \sum_a \pi(a|s_0) Q^{\pi}(s_0, a) \quad \forall s_0 \in \mathcal{S}$$

$$\begin{aligned} \therefore \nabla_{\theta} U^{\pi}(s_0) &= \nabla_{\theta} \left(\sum_a \pi(a|s_0) Q^{\pi}(s_0, a) \right) \\ &= \sum_a \left(\nabla_{\theta} \pi(a|s_0) \right) Q^{\pi}(s_0, a) \end{aligned}$$

$$+ \pi(a|s_0) \nabla_{\theta} Q^{\pi}(s_0, a)$$

$$= \sum_a \left(\nabla_{\theta} \pi(a|s_0) Q^{\pi}(s_0, a) \right. \\ \left. + \pi(a|s_0) \nabla_{\theta} \left(R_{s_0}^a + \gamma \sum_{s', r} p(s', r|s, a) V^{\pi}(s') \right) \right)$$

$$= \sum_a \left(\nabla_{\theta} \pi(a|s_0) Q^{\pi}(s_0, a) \right. \\ \left. + \gamma \pi(a|s_0) \sum_{s', r} p(s', r|s, a) \nabla_{\theta} V^{\pi}(s') \right)$$

Letting $\sum_r p(s', r|s, a) = \bar{p}(s'|s, a)$,

$$\begin{aligned}
 & \downarrow \\
 & = \sum_a \left(\nabla_{\theta} \pi(a/s_0) Q^{\pi}(s_0, a) \right. \\
 & \quad \left. + \gamma \pi(a/s_0) \sum_{s'} \bar{P}(s'|s, a) * \nabla_{\theta} v^{\pi}(s') \right)
 \end{aligned}$$

Unrolling the RHS, we obtain eventually,

$$\begin{aligned}
 & \nabla_{\theta} v^{\pi}(s_0) \\
 & = \sum_x \sum_{k=0}^{\infty} \gamma^k P(s_0 \rightarrow x, k, \pi) \sum_a \nabla_{\theta} \pi(a/x) * Q^{\pi}(x, a)
 \end{aligned}$$

$$\therefore \nabla_{\theta} v^{\pi}(s_0) = \sum_x d_{\pi}(x) \sum_a \nabla_{\theta} \pi(a/x) * Q^{\pi}(x, a)$$

where
$$d_{\pi}(x) = \sum_{k=0}^{\infty} \gamma^k P(s_0 \rightarrow x, k, \pi)$$

Consider the example of parametrized Boltzmann or soft max policies

$$\pi_{\theta}(a|x) = \frac{e^{\theta^T x(s,a)}}{\sum_{b \in \mathcal{A}} e^{\theta^T x(s,b)}}$$

$$\nabla \pi_{\theta}(a|x) = \nabla \left(\frac{e^{\theta^T x(s,a)}}{\sum_{b \in \mathcal{A}} e^{\theta^T x(s,b)}} \right)$$

$$= \sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)} \nabla (e^{\theta^T x(s,a)}) \\ - e^{\theta^T x(s,a)} \sum_{b \in \mathcal{S}} \nabla (e^{\theta^T x(s,b)})$$

$$\left(\sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)} \right)^2$$

$$= \cancel{\sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)}} * e^{\theta^T x(s,a)} \cdot x(s,a)$$

$$\left(\sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)} \right)^2 = \pi_{\theta}(a/s)$$

$$- \frac{e^{\theta^T x(s,a)} \sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)} x(s,b)}{\left(\sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)} \right)^2}$$

$$= \pi_{\theta}(a|s) x(s,a) - \pi_{\theta}(a|s) \sum_{b \in \mathcal{S}} \frac{e^{\theta^T x(s,b)} x(s,b)}{\sum_{b \in \mathcal{S}} e^{\theta^T x(s,b)}} \\ \parallel \\ \pi_{\theta}(b|s)$$

$$= \pi_{\theta}(a|s) \left(x(s,a) - \sum_{b \in \mathcal{S}} \pi_{\theta}(b|s) x(s,b) \right)$$

$$= \frac{V \pi_{\theta}(a|s)}{\pi_{\theta}(a|s)} = \begin{pmatrix} x(s,a) \\ - \sum_{b \in \mathcal{S}} \pi_{\theta}(b|s) x(s,b) \end{pmatrix}$$

Policy Parametrization der Continuous Actions

Example:

Set of actions = $\mathbb{R}$

Actions chosen according to

$N(\mu, \sigma^2)$

$$\text{PDF: } p(x) \triangleq \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ & σ are mean & std. dev.
of $N(\mu, \sigma^2)$.

In this case, we may have a parametrized policy as

$$\pi(a|s, \theta) = \frac{1}{\sigma(s, \theta) \sqrt{2\pi}} e^{-\frac{(a - \mu(s, \theta))^2}{2\sigma(s, \theta)^2}}$$

Here $\mu: S \times \mathbb{R}^{d'} \rightarrow \mathbb{R}$ &

$\sigma: S \times \mathbb{R}^{d'} \rightarrow \mathbb{R}^+$ are

two parametrized function approximators

Let $\theta = (\theta_\mu, \theta_\sigma)^T$

$\downarrow$ approximates μ

$\rightarrow$ approximates σ

Suppose $\mu(s, \theta) \triangleq \theta_\mu^T \underbrace{x_\mu(s)}_{\text{state features}}$

$\& \sigma(s, \theta) \triangleq \exp(\theta_\sigma^T \underbrace{x_\sigma(s)}_{\substack{\hookrightarrow \\ \text{also} \\ \text{state features}}})$

$\text{Exp}(\cdot)$ ensures that $\sigma(s, \theta) > 0$